Homework 0

Marwan Kallal

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1.1 Multipath propagation

One transmitted signal is reflected in some way, so that delayed versions of that signal all arrive at the receiver.

1.2 Doppler Effect

Shift in frequency from what was transmitted, usually caused by velocity, though I think that frequency offsets from clock skew might also be considered Doppler.

1.3 Diversity

Sending data across different frequencies or across time.

1.4 Equalization

Reversing the channel effects to recover a less distorted signal, used to recover signal power in fading channels

1.5 Spread Spectrum

Spreading signal power over time or frequency.

1.6 MIMO

Multi-in multi-out, the use of multiple antenna elements usually to do beamforming

1.7 FDMA, OFDMA, TDMA, CDMA

Frequency/Orthogonal Frequency, Time, Cell Division Multiple Access. Different ways of supporting multiple users on the same frequency bands.

1.8 Frequency reuse

1.9 Co-channel Interference

Interference from signal power in the same frequency band.

1.10 5GNR

Current fifth generation cell standard from 3GPP.